

Topic Module	Groundwater	
	1.	Groundwater Systems

This module introduces fundamental definitions, processes and systems for thinking about groundwater. It encourages you to think about groundwater both on large scales and over long time – the groundwater system – as well as the impacts that humans can have on its natural state. This module is a foundation for later topics on groundwater well design, and advanced courses on groundwater modelling.

Why is groundwater important?

Groundwater is used as a potable drinking water supply, for irrigation and for industrial processes. It is especially important for the Canterbury region, which accounts for about 90% of New Zealand's groundwater use.

Groundwater is an essential part of the hydrological cycle. Aquifers provide a temporary home for infiltrating precipitation (or snowmelt), and the slow release as baseflow helps maintain stream and lake levels during dry seasons. As groundwater undertakes its long journey beneath the surface, toxins and pathogens are filtered out (although some minerals may be acquired along the way).

There are many consequences for mismanaging groundwater. If contamination is allowed to occur, the resource may become unusable for many decades, or require expensive treatment. If individuals pump groundwater too close to one another, they may end up impacting each other's wells. If aquifers are over allocated, then the water table can drop and streams that rely on baseflow can run dry. For coastal aquifers, a reversal in the direction of groundwater flow can draw seawater into the resource.

In New Zealand, and many other countries, the groundwater system is tightly regulated. Groundwater wells must be designed and installed to a high standard to prevent contamination and interference with neighbors. At the regional scale, aquifer systems are monitored for quantity and quality, with groundwater abstraction managed through environmental consent. Finally, computer models are used to predict the sustainability of a groundwater system for different use scenarios, or in response to long-term environmental forcings like climate change.

Where is groundwater?

Groundwater occupies the open spaces underground. It fills pores (in porous rock), fractures (in fractured rock) and open caverns in special rock formations that prone to dissolution, e.g., limestones (Figure 1).

Close to the surface, the open spaces are filled by a mixture of water and air. This is called the unsaturated, or vadose zone. The saturation measures the fraction of open space that is filled by water.

Eventually, there is a depth where all the open space (also called porosity) is filled by water and no air remains. This is the saturated, or phreatic zone, and the depth that it begins is called the water table (Figure 2).

Here, we focus exclusively on the saturated zone. The flow of water through the unsaturated zone is still important, particularly for infiltration, but it is also much more complicated.

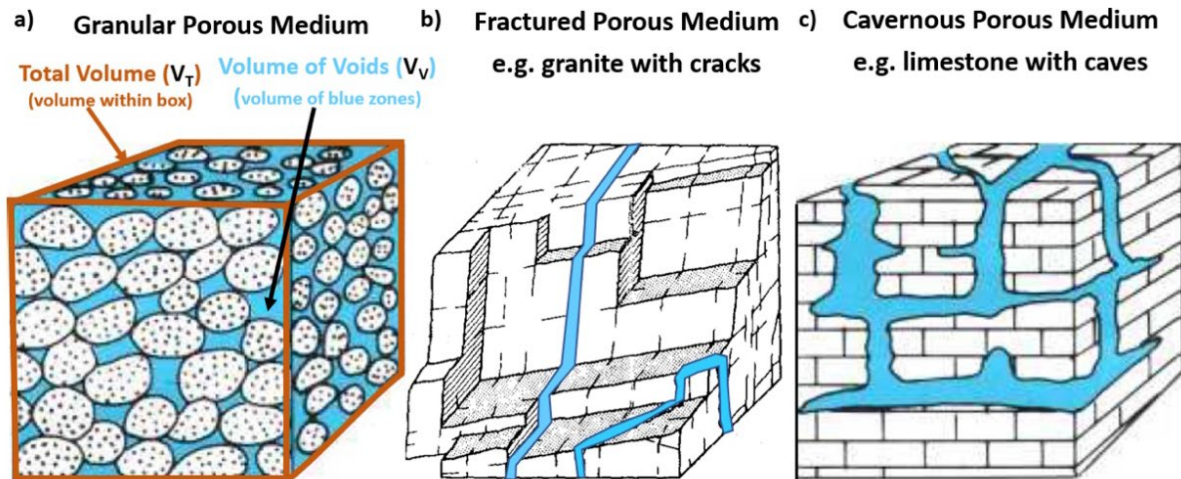


Figure 1. Groundwater occupies the open space (porosity) in geologic formations. This can be (a) the space between rock grains, (b) in open fractures, or (c) in caverns formed from dissolution (from Woessner and Poeter, 2020).

What is a groundwater system?

To understand what is important in groundwater systems, it can help to think about the different journeys a single drop of water could take (Figure 2).

Most groundwater begins as precipitation, either as rain directly falling onto and then infiltrating the land surface, or as snow that accumulates temporarily and then later melts.

A drop of water entering at the surface – recharge – infiltrates through the soil column and eventually arrives at the water table. Here, it flows under the force of gravity through a connected framework of open pores. The ease with which a drop can squeeze from one pore through to the next is the hydraulic conductivity of the rock (conceptually, it is quite similar to electrical and thermal conductivity).

Rocks with high porosity and high conductivity are called aquifers: they are often a good source of groundwater. Rocks with low porosity or low conductivity are called aquitards: while not a source of groundwater, they provide important protection to aquifers from surface contamination.

An aquifer that sits below an aquitard is called a confined aquifer: the top of the aquifer is the bottom of the aquitard. If there is no aquitard over the top, it is an unconfined aquifer: the top of the aquifer is the water table.

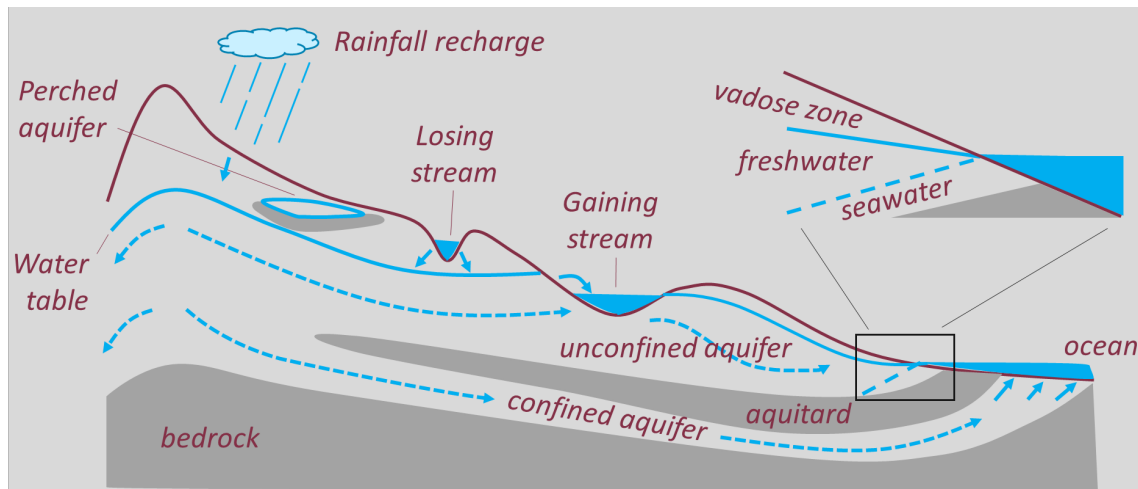


Figure 2. Schematic depiction of a groundwater system, tracking the journey of water through the subsurface: from rainfall to ocean.

After travelling through an aquifer for a while, a drop of water may eventually exit back to the surface. For example, if the water table is higher than the bottom of a stream, then water can exit directly into the stream. This is called baseflow and the stream is termed a gaining stream. The opposite can also occur: if the water table is below the bottom of a stream, then water can exit the stream into the groundwater system. This is a losing stream. A stream can be gaining and losing along different reaches, e.g., the Waimakariri River.

Other drops of water will exit an aquifer directly into the ocean. Other times, an interface between the fresh groundwater and saline seawater will form *inside* the aquifer. Seawater is heavier and therefore infiltrates the lower part of the aquifer.

Finally, for very special aquitard shapes, a perched aquifer may form, where infiltrating water becomes trapped and unable to flow out. Because they tend to be small, perched aquifers are vulnerable to overuse.

How best to think about groundwater?

When thinking about groundwater dynamics in a particular area, it is useful to have a conceptual model of the groundwater system. This might be obtained from consultant reports, from published academic articles, through discussions with scientists, or developed yourself from the available evidence.

The conceptual model is a description of the geometry, processes and quantities for a specific groundwater system. It includes the depths, thicknesses and extents of aquifers and aquitards. It includes the location and quantity of water you think is entering the system (recharge) as well as where it leaves (discharge). Where appropriate, it should include any connected surface water bodies, and inferred flow pathways.

A conceptual model is a best guess based on the available evidence. That means, if you new evidence is uncovered, it is entirely appropriate to update your conceptual model. It is typically expressed as a simplified, 2D cross-sectionional drawing.

Conceptual model examples:

One of the earliest and best-known conceptual models was developed by NH Darton of the United States Geological Survey at the turn of the 20th century. The goal was to understand the origins of a huge artesian groundwater source that underlay much of South Dakota. The water was clearly not derived from local rainfall. Instead, Darton proposed that water entered the aquifers many kilometers to the west in the Black Hills, where it travelled through the Dakota Sandstone and was confined above by cretaceous shales.

In Canterbury, groundwater is contained in a series of interlayered gravel aquifers that are separated by clay aquitards. Groundwater enters the system much further inland, flowing beneath the Canterbury Plains and eventually exiting up to 40 km offshore. Some leakage between the aquifers is presumed to occur across the aquitards.

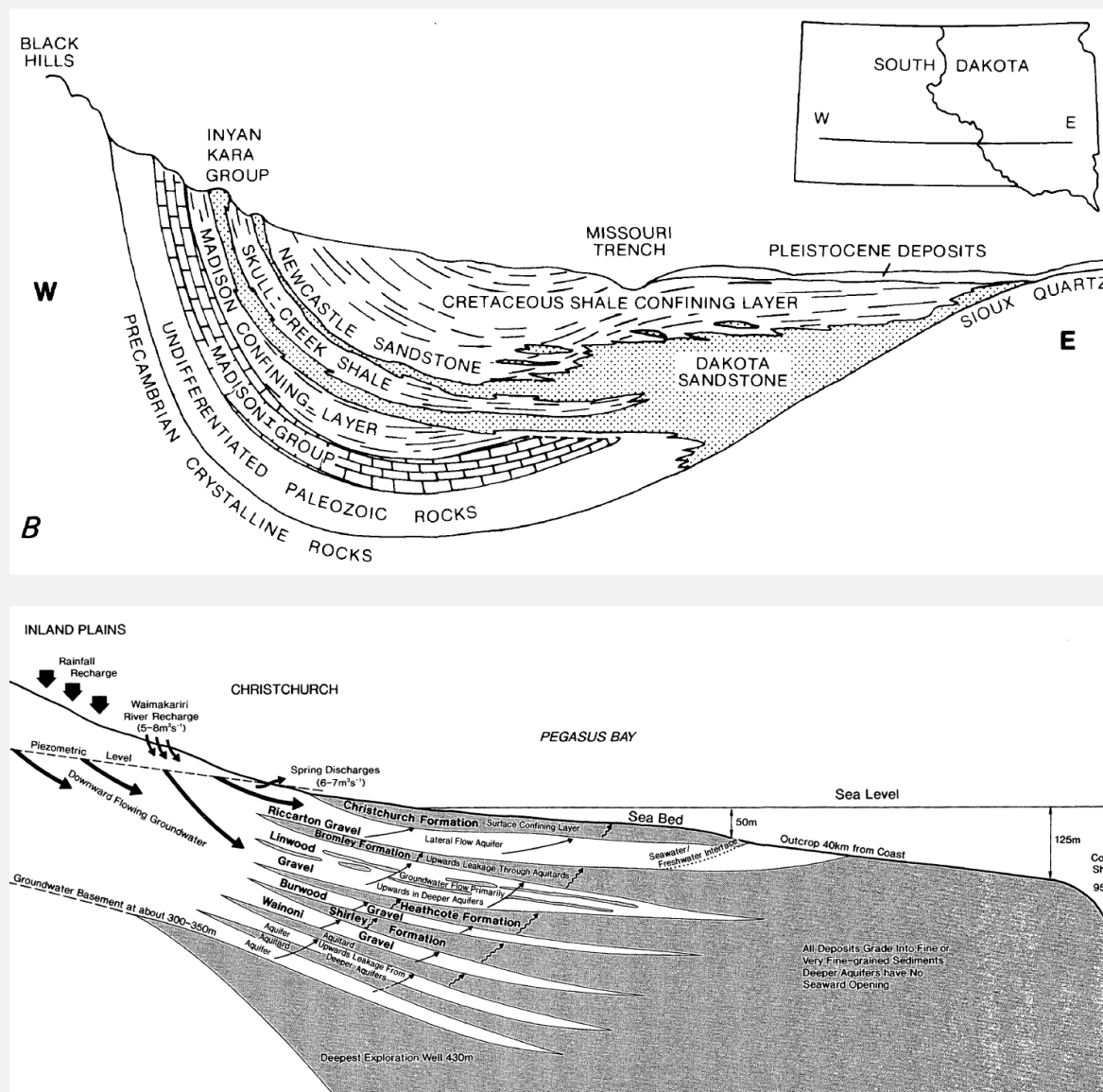


Figure 3. (Top) Conceptual model of the Dakota Plains confined groundwater system (Bredehoeft et al. 1983). (Bottom) Conceptual model of the Canterbury Plains groundwater system (Brown et al., 1995).

What influence do rocks have on groundwater?

Hydrogeology is the study of the properties and distributions of rocks that are relevant to groundwater systems. A hydrogeologist is an expert at studying, understanding and modelling groundwater systems.

An aquifer is any rock formation that contains a sufficient quantity of accessible water to be useful as a groundwater resource. Aquifers in Canterbury are comprised of thick sections of buried gravel or sand (Figure 4), which were laid down many thousands of years ago by flooding of our meandering braided rivers. In Auckland, the aquifers are fractured lava flows from old erupted volcanoes. In Nelson and Golden Bay, they are great limestone deposits whose fractures have been slowly dissolved and hollowed out by percolating rainwater.

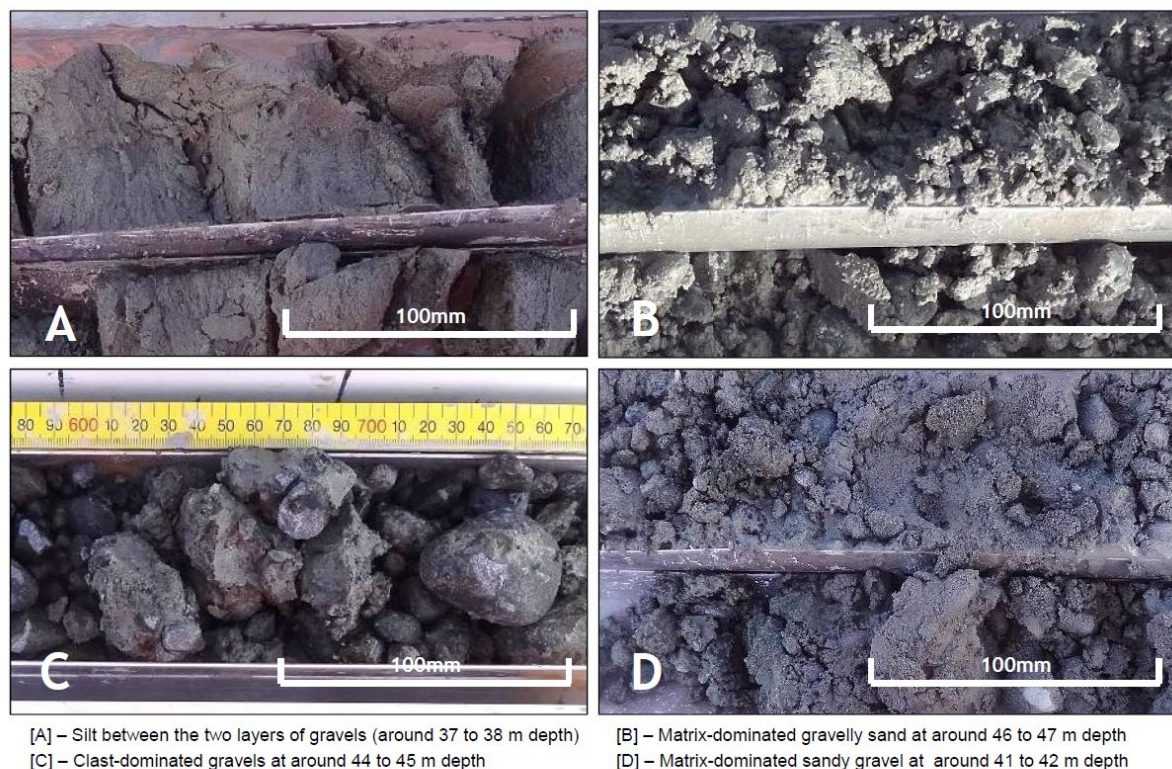


Figure 4. Drill cuttings recovery showing different material compositions from a borehole near Bridge Street, Christchurch (Green et al., 2015). The large grain size makes it easy for water to pass through these materials.

In contrast, an aquitard is any rock or formation that strongly resists the flow of water. In Canterbury, these are clays, silts or peats that were deposited when the area was covered by coastal swamps (Figure 5). The growth and retreat of glaciers over climate cycles can grind up rocks into a very fine till which forms an effective barrier to water. In geothermal fields, exposure of rock to high temperature water can cause rock to slowly alter to an impermeable clay. In other places, metamorphic shales provide low permeability aquitards.



Figure 5. Examples of aquitard material (left) clay cored from a drillhole and (right) fine-grained till deposited during glaciation. In both cases, the small grain-size makes it difficult for water to pass through the formation.

The juxtaposition of aquifers against aquitards is the foundation of archetypal groundwater systems. For example, the beginning of an aquitard marks the base of an unconfined aquifers – therefore, aquitard depth controls an aquifer's saturated thickness. Aquitards demarcate the top and bottom of confined aquifers, although they may leak in places providing a vertical connection with overlying aquifers. A seepage face (Figure 6) occurs when an aquifer-aquitard contact outcrops at a cliff face. The groundwater, having no where else to go, seeps out the side of the cliff.



Figure 6. Example of a seepage face, where an aquifer is laterally exposed at a cliff face. In this case, a limestone aquifer overlying a shale aquitard.

As water passes through an aquifer, it can become altered through contact with the rock. Groundwater can contain high concentrations of dissolved ions that contribute to the water's hardness and may require treatment before use. Studying the geochemistry of groundwater can tell you where it has come from, how long it has been underground, and what rocks it has been in contact with.

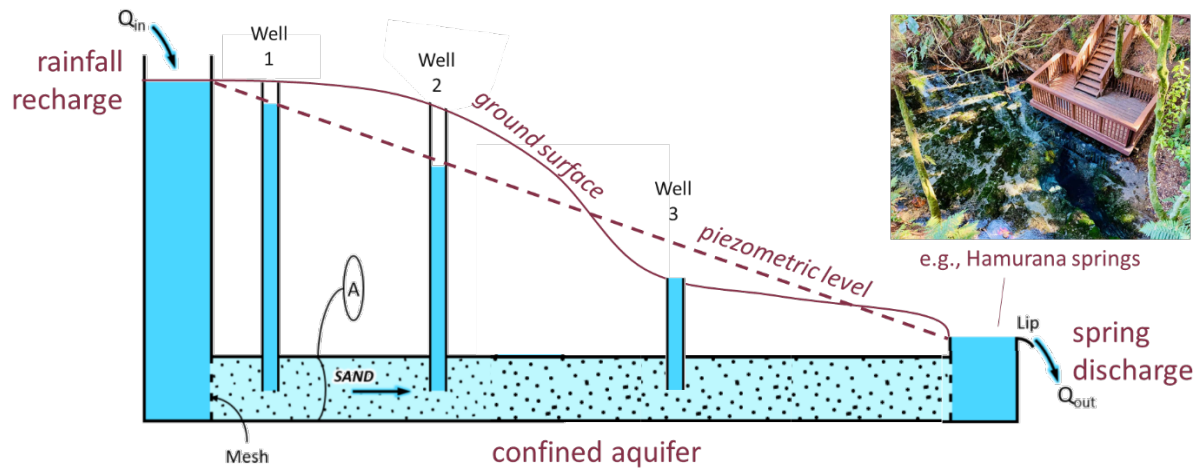


Figure 7. An artesian spring (e.g., Hamurana springs near Rotorua) conceptualized as sand-piezometer experiment. Water enters at the left (rainfall) and flows through a sand-filled tube (confined aquifer), losing pressure/energy as it goes. Water flows out of the tube at the right (an artesian spring). Illustrative ground (solid) and piezometric levels (dashed) are shown, along with the expected water level response in several wells.

Springs

In some locations, groundwater naturally wells up out of the ground: this is called a *spring*. Classic examples in New Zealand are the Hamurana Springs on the north side of Lake Rotorua, and Te Waikoropupū Springs in the Tasman District. In both cases, the upwelling water originates as rainfall in the hills above the spring, before taking an underground journey and returning to the surface.

Springs occur where the natural water level – if you like, the level of the water table – sits above the ground surface. We say the groundwater is *artesian* and this means that it wants to be flowing out of the ground (although it may be prevented from doing so if confined by an aquitard.) One way to think about a spring system is as a sand-piezometer experiment. The high-water level maintained by rainfall in the hills is transmitted as a pressure force through the aquifer that later drives water upward (Figure 7).

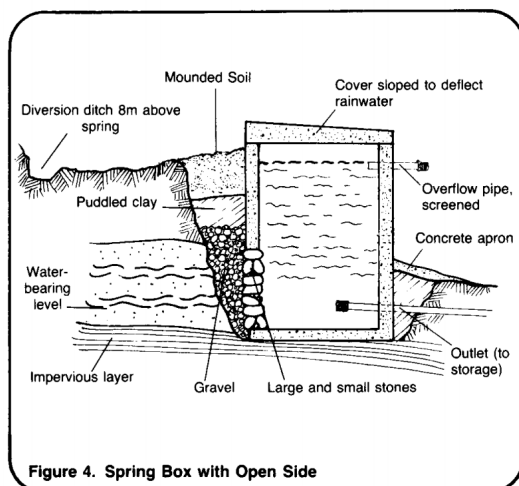


Figure 4. Spring Box with Open Side

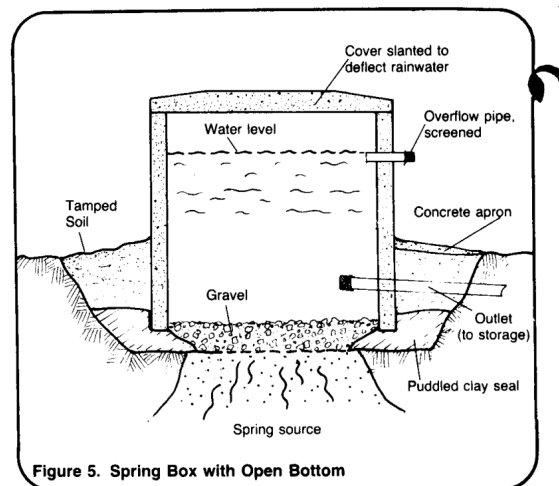


Figure 5. Spring Box with Open Bottom

Figure 8. Engineering of a spring water source for (left) laterally exiting water, and (right) vertical upwelling.

Springs were some of the earliest groundwater resources exploited for drinking water. To ensure that the water supply is not contaminated, it is important to install a housing system around the spring (Figure 8). This should include large stones or gravel to allow entry of the spring water, and secure container to provide buffer storage with overflow and outflow pipes. Most importantly, the water supply should be protected with low conductivity clays or a concrete apron that prevent any surface contamination leaching down to the spring water. The last thing you want is to lose you water supply because someone accidentally spilled a cup of E. coli at the spring.

Much later, people realized they could create their own springs by digging a deep hole – a well – down a groundwater aquifer. If the aquifer was artesian, the well would flow naturally, but if not, some sort of pumping or bailing would be required. The groundwater well is the fundamental unit of groundwater infrastructure.

Groundwater well

A well is first and foremost, a long hole that is dug or, more commonly, drilled deep into the ground (Figure 7). Groundwater wells can range from a few metres to several hundred metres depth depending on the target aquifer. Geothermal wells used for electricity production can be several kilometres deep, and the Deepwater Horizon well – whose blowout caused an oil spill in the Gulf of Mexico – was over 10 km deep (in sea water over 1 km deep).

Once the well has been drilled, improvements are made to enhance the quality of water produced. Casing – an outer annulus pipe – is lowered into the borehole and grouted in place. Its purpose is to prevent water from unwanted aquifers from entering the borehole. A screen (an outer pipe with holes) is lowered below the casing at the depth of the target aquifer. Its purpose is to create a barrier to the groundwater system that allows water to flow into the well, while filtering out large particles. When we say that a well is “screened” in a particular aquifer, we are indicating which formation water is being extracted from.

Groundwater in the well will only rise to the piezometric level. Therefore, a pump, usually submerged, is used to lift the water to the surface so that it can enter the distribution network.

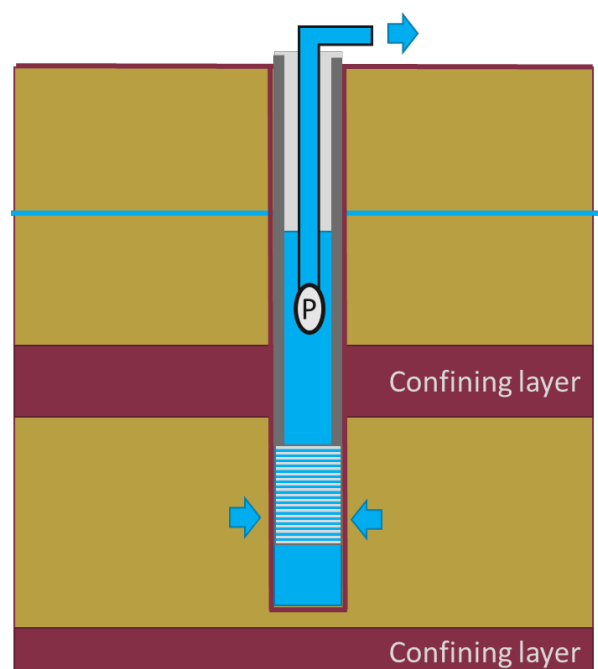


Figure 9. Schematic illustration of a groundwater well with casing (grey) down to the confined aquifer in which it is screened (horizontal hatching). Water only rises in the well to the piezometric level (blue line). It must be lifted out by a submerged pump.

Groundwater head

Much like the flow of water through pipes, we can think about groundwater as being characterized by either a pressure or a head. Groundwater head in an aquifer is the level to which the water would rise in an infinite straw that is open to just that aquifer (Figure 10). Head is measured relative to a datum, for example, the ground level at your current location, or sea level.

Pressure and head are related by the equation:

$$p(z) = \rho g(z + h) \quad (1)$$

Where,

p = Pressure (Pa)

ρ = Density (kg/m³)

z = Depth below the water surface (m)

h = Head (m)

The first term in (1) is the hydrostatic pressure due to the weight of water, and the second term is the additional pressure reflected as non-zero head.

In an aquifer where water can flow easily, the head will be the same at all depths but the pressure will increase hydrostatically with depth. Head can be different for overlying aquifers in a sequence providing they are separated by aquitards. The aquitard maintains a head difference by preventing the flow of water from the low-head aquifer to the high-head one.

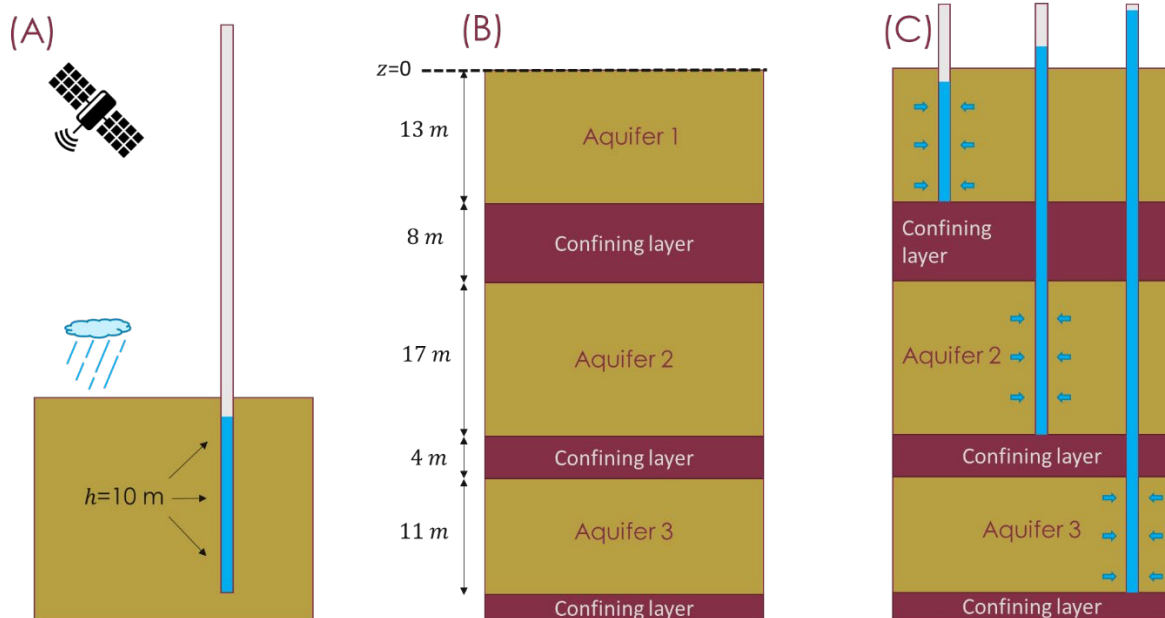


Figure 10. Piezometric surfaces with depth. (A) Schematic illustration of piezometric surface as the height to which water would rise in a straw that penetrates the aquifer of interest. (B) Aquifer setup for Example Problem 1. (C) Illustration of a wells penetrating a three-aquifer system, each with different heads. The head difference can be maintained because the confining layers that separate the aquifers prevent flow of water that would otherwise equilibrate head (pressure).

Example Problem 1:

Consider the aquifer system shown in Figure 10B and answer the following questions.

1. The pressure at the top of Aquifer 2 is 0.226 MPa, what is the head?
2. The head in Aquifer 3 is 5 m, what is the pressure at the mid-point?
3. The pressure at the bottom of Aquifer 1 is 0.118 MPa, what is the head?
4. Where is the water table?
5. Head is “everywhere” the same in a static column of water. Yet it is *not* the same for these three aquifers. Explain how this can be the case.

Piezometric surface

Head varies not just with depth, but also laterally within an aquifer. By measuring head at many different locations within an aquifer, we can guess at the values in between to create contours of constant head (Figure 11A). Interpreting head more literally as an elevation (that water would rise to), these contours define a piezometric surface.

In an unconfined aquifer, the piezometric surface is the same as the water table. If it is higher than the ground surface, then this usually corresponds to the water surface in a wetland.

In a confined aquifer, the piezometric surface is *hypothetical* and usually sits above the aquifer itself. If it is higher than ground level, then the aquifer is said to be artesian. A well drilled into it would flow without any pumping.

The piezometric surface is usually highest in inland regions where a lot of rainfall is recharging the system. From there, it will tend to decrease with elevation, reaching sea level near to the coast. However, there are notable exceptions, with an elevated piezometric surface extending many kilometers offshore from Christchurch for some deep aquifers (Figure 3).

Groundwater flows in the direction of steepest descent down the piezometric surface. In other words, flow arrows are perpendicular to contours of constant head (Figure 11A).

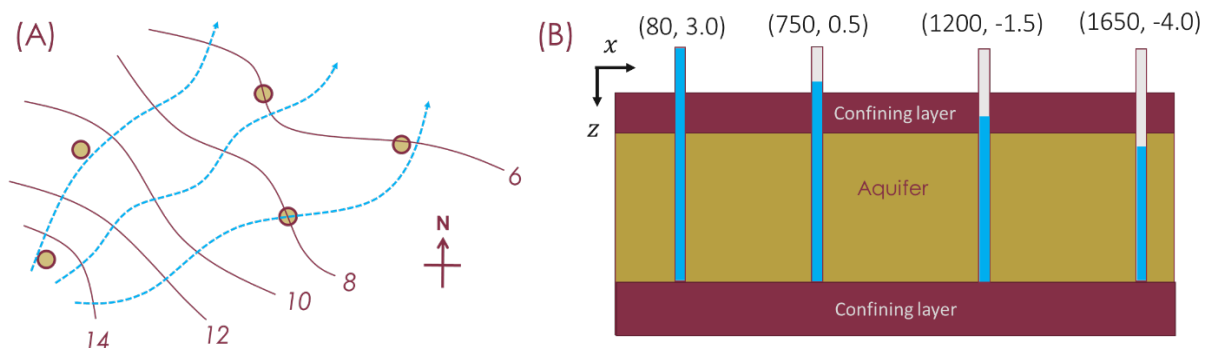


Figure 11. Piezometric surfaces horizontally. (A) Schematic illustration of piezometers (circles), lines of constant head (brown) and their perpendicular flow lines (blue). (B) Aquifer setup for Example Problem 2. Bracketed quantities give the x coordinate of the piezometer, and then the measured head value, both in m.

Example Problem 2:

Consider the aquifer system shown in Figure 11B and answer the following questions.

1. Compute the linear hydraulic gradients between the three pairs of adjacent wells.
2. Sketch the piezometric surface.
3. Compute the hydraulic gradient between the farthest two wells. Comment on how this differs from the gradients in question 1.
4. Thus argue: what is the value of more data? what are the limits of inference for piezometric surfaces?

Darcy's Law

Water flows down the piezometric surface, but at what rate? We use Darcy's Law to describe the volume flow rate of water (in m³/s) in terms of properties of the rock and the flow field (Figure 12A). Formally:

$$Q = -KA \frac{dh}{dx} \quad (2)$$

Where,

Q = volume flow rate (m³/s)

K = hydraulic conductivity (m/s)

A = area across which flow occurs (m²)

$\frac{dh}{dx}$ = hydraulic gradient (m/m or dimensionless)

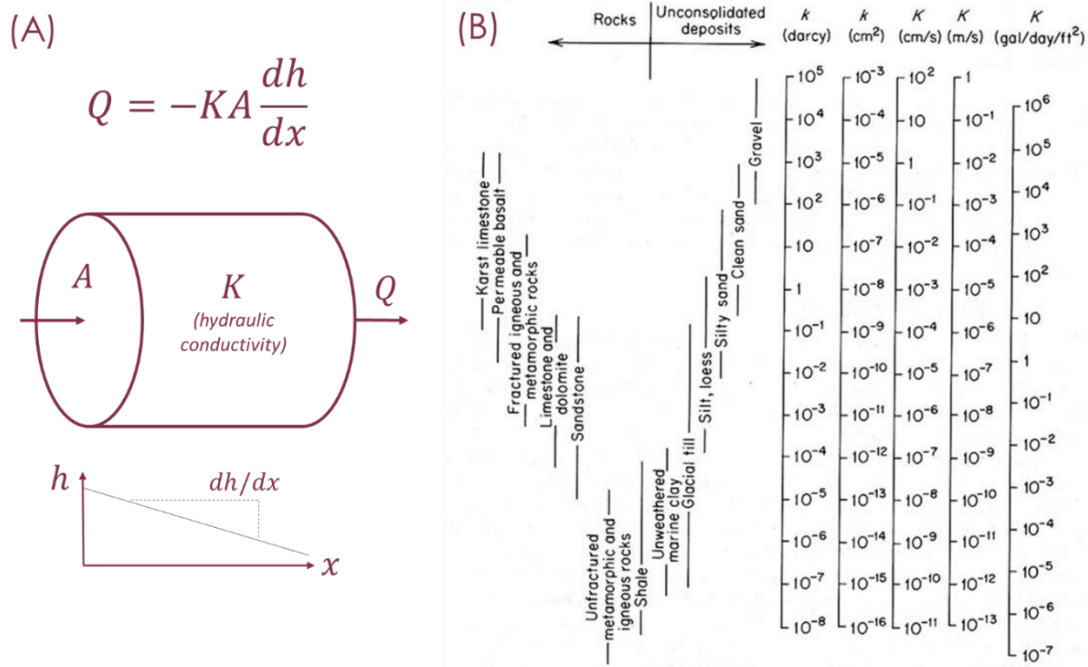


Figure 12. Darcy's law and hydraulic conductivity, K . (A) Schematic illustration of Darcy flow through a cylindrical control volume. (B) Range of hydraulic conductivity by rock type. Note the log-scale used for measuring K .

The hydraulic conductivity, K , varies over many orders of magnitude depending on the type of rock. For instance, gravels are the most conductive aquifer rock, with conductivities between 10^{-3} and 1 m/s, whereas aquitards like clay range from 10^{-10} and to 10^{-7} m/s (Figure 12B).

Armed with this equation, we can calculate the amount of water passing through a given area of the aquifer. Sometimes we'll use a slightly different version of Darcy's Law, $q = -K \frac{dh}{dx}$, where q is the specific discharge (volume flow rate per unit area). The hydraulic gradient is usually calculated as a finite difference between two known measurements of head.

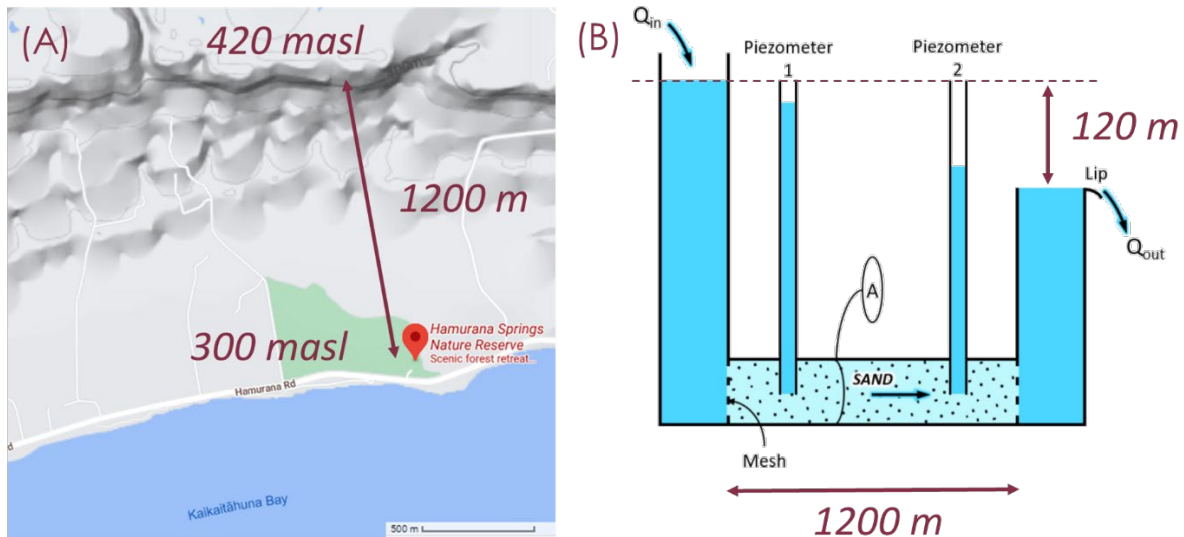


Figure 13. (A) Hamurana springs conceptualized as (B) a sand-piezometer experiment. Marked dimensions are relevant to Example Problem 3.

When solving groundwater problems, it is often convenient to assume steady state conditions. If the groundwater system is not changing over time, then the amount of water flowing into a volume has to be balanced by the volume flowing out of it. Apply this concept to solve the problems below:

Example Problem 3:

Consider the Hamurana Springs problem shown in Figure 13 and answer the following questions.

1. If there is an open, connected system from the overhead bluffs to the springs, what are the maximum and minimum possible hydraulic gradients connecting the recharge region to the spring?
2. For the maximum estimate of the hydraulic gradient, and a spring discharge of $4.63 \times 10^{-2} \text{ m}^3/\text{s}$, calculate the product KA .
3. What is required to determine K or A separately?
4. If the main aquifer transmitting spring recharge was a gravel, estimate a range for A .
5. Note, A is the area of *open pore space* that groundwater can move through. If the gravel is 40% open pore space, and 60% solid rock, what is the *effective area* across which flow occurs?

Example Problem 4:

The flow system in Figure 14A has a discontinuity that temporarily reduces aquifer thickness and permanently changes hydraulic conductivity (but pore space is the same).

1. By equating the volume flow rate between wells 1 and 2, and between wells 2 and 3, find the possible range of K for Aquifer 2 (assume a constant out-of-plane dimension).
2. A new well 4 is to be drilled 800 m downstream from well 3. What is the expected range of head in this well?
3. Explain how drilling well 4 will help to constrain the size of the flow constriction.

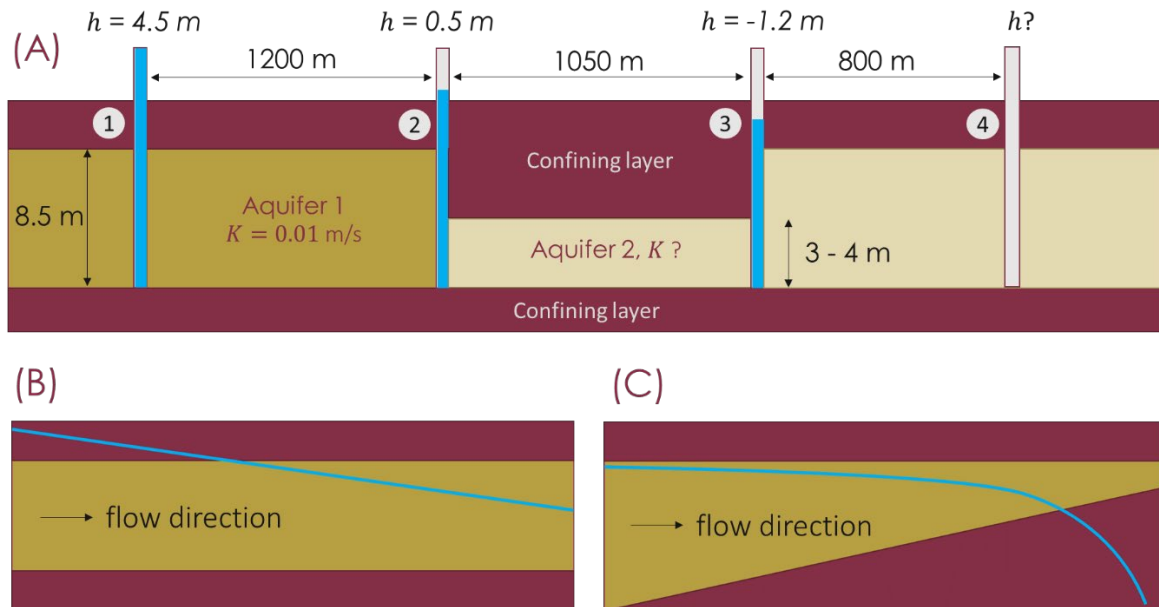


Figure 14. Flow through constricted aquifers. (A) Setup and quantities for Example Problem 4. (B) A linear hydraulic gradient for steady flow in an aquifer of constant thickness and conductivity, and (C) a logarithmic gradient in an aquifer with a linear pinch out.

Example Problem 5:

The flow systems in Figure 14B and C are in steady state.

1. If K and A are constant, show that the hydraulic gradient must be linear.
2. If K is constant and A is decreasing linearly (a pinch out), show that the hydraulic gradient is logarithmic.

When applying Darcy's Law to calculate the flow in an aquifer, we will often assume that cross-sectional flow area (A) is constant. Thus, in many confined aquifers, the hydraulic gradient is linear (at least locally).

For an unconfined aquifer, the cross-sectional flow area is the saturated thickness of the aquifer. This, in turn, is given by the head, which is the water table for an unconfined system. As a result, under steady conditions, Darcy's Law leads to a parabolic/square root head profile (water table) in the unconfined aquifer (Figure 15). This is called Dupuit-Forchheimer flow.

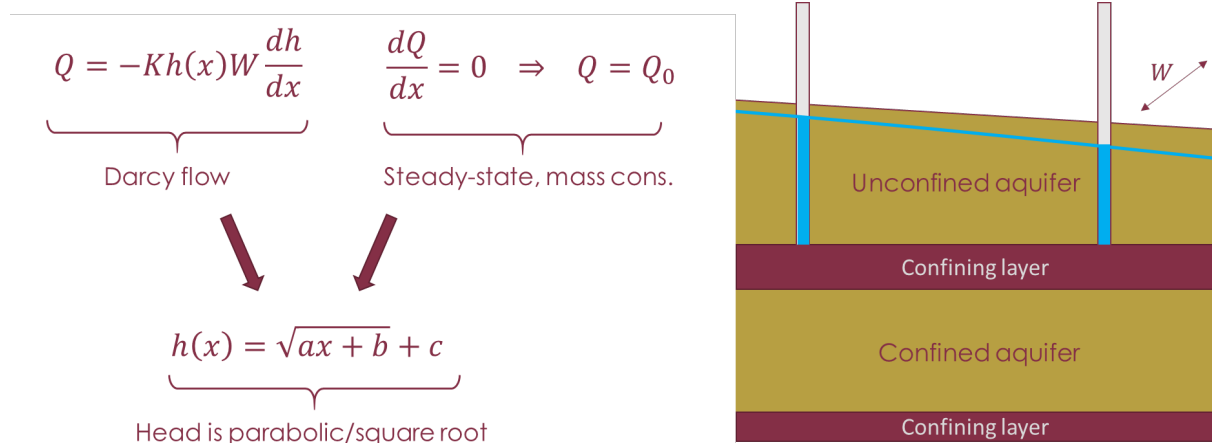


Figure 15. Dupuit-Forchheimer flow. In an unconfined aquifer, where a declining water table reduces the saturated thickness of the aquifer, the steady-state head solution is parabolic/square root.

What happens when we use groundwater?

So far, we have mainly thought about the groundwater system as being in a steady state. Water is certainly moving through it, but there is a natural equilibrium where flows in and out of any volume always balance, and the flows themselves don't change over time.

Drilling a groundwater well and pumping water out of an aquifer disturbs this equilibrium. A different one may later be established, but, until then, flow rates will change over time and there will be imbalanced flows into and out of volumes. There are two important concepts for understanding this disequilibrium: drawdown and storativity.

Aquifer storage

When we drill and then pump a well, we are removing a volume of groundwater from the aquifer. The result is a decrease in pressure, or head, in the aquifer. The drawdown is the change in the piezometric surface, or head, compared to its pre-pumped state, i.e., $\Delta h = h - h_0$. Around a well, this tends to assume a circular shape called the cone of depression.

Pulling a larger volume of water out from underground will cause a larger drawdown. Storativity is defined as the volume of water that must be extracted to induce one unit of drawdown evenly across one unit area of the aquifer:

$$S = \frac{\Delta V}{A\Delta h} \quad (3)$$

Where,

S = storativity (dimensionless)

ΔV = volume of water removed (m^3)

A = area across which drawdown occurs (m^2)

Δh = drawdown (m)

Field tests have shown that storativity is orders of magnitude larger in unconfined aquifers than confined. That is, removing one cubic meter of water induces a much smaller decrease of the water table, than it does a decrease of a confined aquifer's piezometric surface.

The reason has to do with *how* the aquifer releases water. For an unconfined aquifer, the water is drained from the pore space. Storativity is equal to the drainable porosity, also called the specific yield, S_y . It has values between 0.05 and 0.5.

For a confined aquifer, water is *released* through elastic decompression of the rock and water itself. It may be helpful to think about the reverse process: injection. To introduce more water to a confined aquifer – a fixed volume with no water table that can increase – we need to compress the existing water and rock grains into a smaller space.

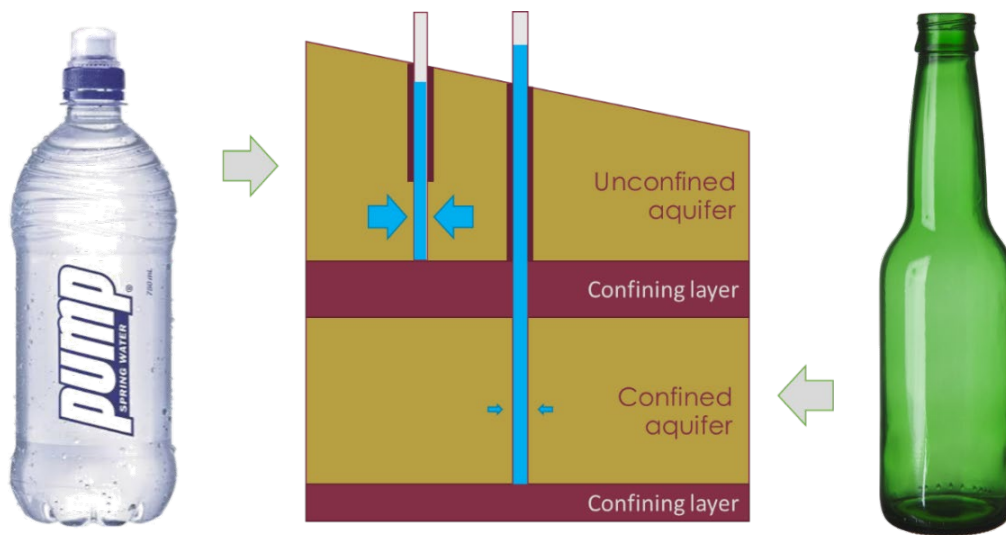


Figure 16. The difference between pumping water from unconfined and confined aquifers and a bit like the difference between drinking from plastic and glass bottles. For an unconfined aquifer (plastic bottle), only a small change in head (suction pressure) is required to produce a lot of water. For a confined aquifer (glass bottle), a large change in head is required to produce a small amount of water. The difference is storativity.

The elastic storativity is proportional to aquifer thickness, b , and a compressibility quantity called the specific storage, $S_s = \rho g(\beta_r + n\beta_w)$ where ρ is water density, n is porosity and β_r and β_w are rock and water compressibility. S_s is usually small, between 10^{-6} and 10^{-4} .

What this means is that it is harder to pump water out of a confined aquifer than an unconfined one. Think about drinking from a bottle (Figure 16). In a similar way, it is difficult to suck a fluid from a glass bottle while maintaining a perfect seal on the glass neck. Allowing air into the bottle and letting the water level lower, makes it much easier to remove the liquid. Finally, more compressible plastic bottles (more compressible rocks) are easier to extract water from.

Remember, to calculate storativity, first decide the nature of the aquifer. If it is a confined aquifer, remember to multiply by the thickness.

$$S = \begin{cases} S_y, & \text{unconfined aquifer} \\ bS_s, & \text{confined aquifer} \end{cases} \quad (4)$$

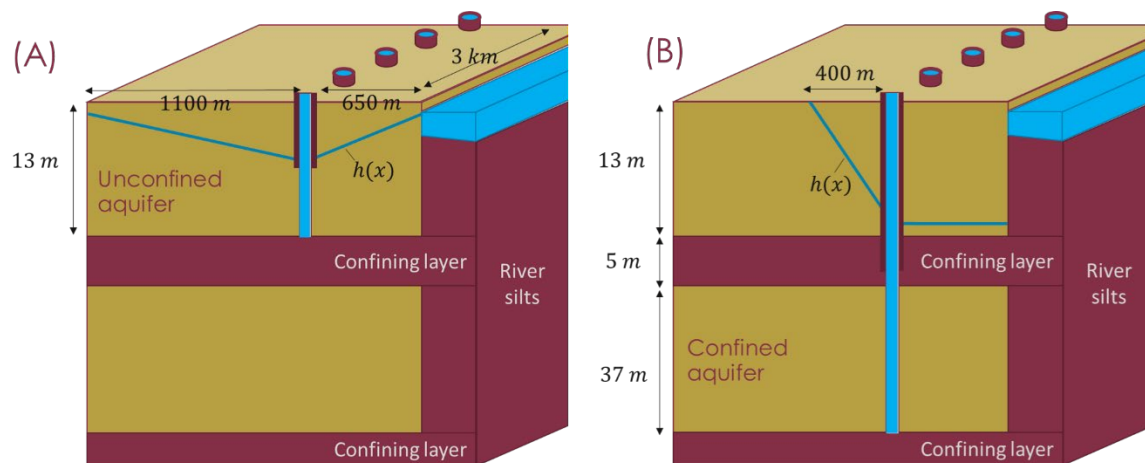


Figure 17. Bank filtration Example Problem. A battery of groundwater wells parallels a river. Producing from these wells draws water from the river, through the rock, and into the wells, cleaning the water in the process. (A) An example showing production from an unconfined aquifer connected to the river. (B) An example showing production from a confined aquifer that is laterally constrained by the river.

Bank Filtration:

A bank filtration scheme operates a battery of wells that are set back 650 m along a 3 km stretch of a river. The wells target an unconfined sand aquifer (Figure 17A).

After, 10 years and 3.3 km^3 extraction, the average water level in the wells has dropped 5 m.

Monitoring wells show the water table is unchanged 1.1 km away.

1. What choice have we made for the level of the water table adjacent to the river?
2. The main aquifer is a sand with a specific yield of 0.2. How much water has been released from storage?
3. Why is this not the same as the extracted water? What could account for the shortfall?
4. Prior to development, the average annual flow rate in the river was about $100 \text{ m}^3/\text{s}$. Is this scheme likely to have substantially affected the flow?

Due to environmental concerns, the wells were deepened to target a thick, confined gravel (Figure 17B). The gravels are also laterally confined by a thick deposit of river silts and sediments.

In 6 months, head declined 12 m at wells, recovering to background levels 400 m away.

5. What choice have we made in conceptualising head decline between the wells and the river?
6. If S_s of the gravel is 2.3×10^{-5} , how much water has been released from storage?

Groundwater Capture

A conceptual model of a groundwater system should summarize all the flows into and out of an aquifer. For example, Figure 18A shows a simplified aquifer with a steadily dropping piezometric surface from left to right. This means water is flowing from left to right, including at the boundaries.

Recharges to the system include precipitation and water flowing in through the lefthand boundary from a neighboring aquifer volume. Discharges include loss to a stream (it sits under the piezometric surface) and water exiting through the righthand boundary to another aquifer volume.

Under steady conditions, the total recharge, R_0 is balanced by the total discharge, D_0 , and there is no change in total water volume, i.e., $\frac{dV}{dt} = 0$.

A well is drilled and water is abstracted at rate, Q . At that instant, before recharge and discharge can respond, there must be disequilibrium, i.e., $\frac{dV}{dt} = -Q$.

Over time, a new water table will be established, reflecting drawdown around the well (Figure 18B). Because the hydraulic gradients are different, recharges and discharges may be increased (LH boundary), decreased (RH boundary), or reversed (baseflow). But a new equilibrium is eventually established, with $\frac{dV}{dt} = 0$ once again.

The changes in recharge and discharge, induced in the groundwater system due to pumping, is called groundwater capture. In this case, $\text{capture} = \Delta R - \Delta D$ and it is equal to pumping, Q .

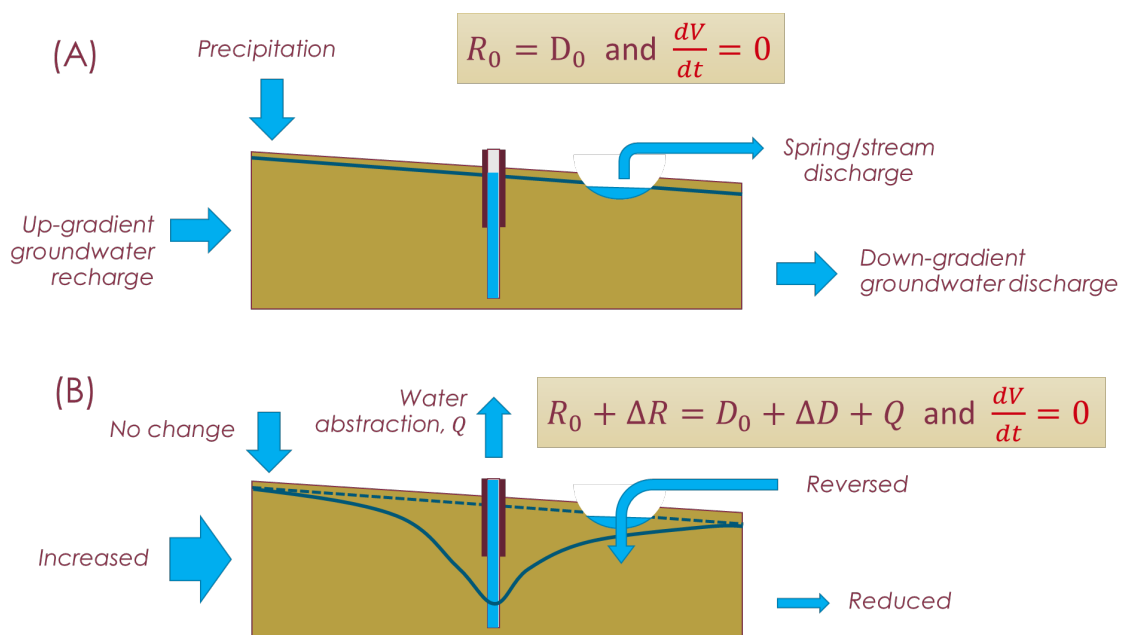


Figure 18. Schematic illustration of the effects of pumping on the natural state of a groundwater system. (A) The system when no pumping is present: recharges balance discharge. (B) The system when abstraction is present: recharge still balances discharge, although the size or direction of individual flows may change.

Example Problem 6:

Consider the aquifer system shown in Figure 19. A well is to be drilled into Aquifer 2 at the location marked X.

1. What is the natural condition of this well?
2. Under heavy pumping, redraw the piezometric surface.
3. Identify and describe sources of capture.
4. Comment on the possible movement of the seawater/freshwater interface.

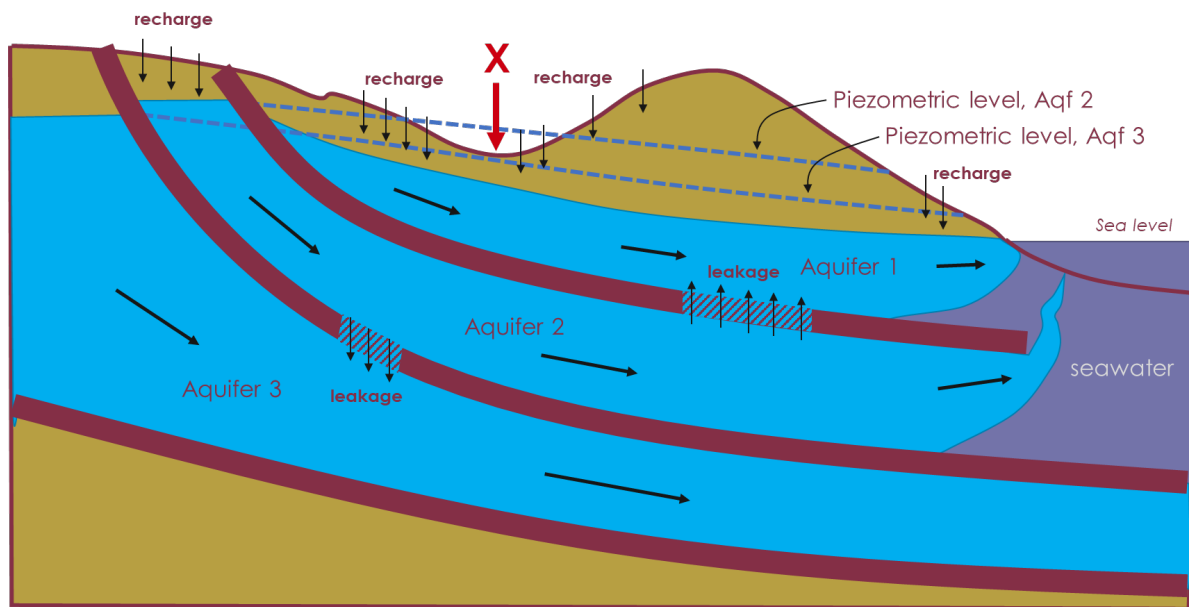


Figure 19. Conceptual model of a multi-aquifer groundwater system with leakage, adapted from Bear (2012). See Example Problem 6.

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